

When a Ride Is the Missing Treatment

Transportation Barriers and Acute-Care Patient Journeys at Stormont Vail Health (2022–2025)

Team 13 — DataFest 2026, Wesleyan University

April 19, 2026

Question

Using the longitudinal EHR data, I asked: *Do patients who report a transportation barrier experience measurably different healthcare journeys than otherwise similar patients?* If yes, this is an upstream actionable signal — a social determinant the health system can act on with shuttle programs, tele-visits, or case management.

Data and Methods

I used the full DataFest 2026 release from Stormont Vail Health: **7,675,801 encounters** for **947,685 patients**, 2022-01-01 through 2025-12-31, joined to diagnoses, departments, providers, and the 12-domain social-determinants (SDOH) questionnaire. I ingested the seven CSV files into a local DuckDB database for columnar queries, parsed `Date + TimeOfDayKey` into an encounter timestamp, and built a patient-level analytic table with encounter counts, ED/inpatient/observation flags, median gap-days between consecutive encounters, and merged demographics.

From the *Transportation Needs* SDOH domain I derived a three-level exposure:

- **Barrier:** patient ever answered *Yes* to either transportation question (**n = 2,986**)
- **No barrier:** ever answered *No*, never *Yes* (**n = 55,653**)
- **Declined / unable:** (**n = 1,438**)

Non-screened patients (**299,699**) were kept as context but excluded from comparative tests. Headline outcomes: ED visits per person-year, inpatient admits per person-year, probability of any ED / any inpatient visit, and a diagnosis-anchored check: % with any ED visit within 180 days after first encounter for I10, E11, N18, or I48.

Key Findings

1. **Transport-barrier patients use 3–4x more acute care.** They average **1.94 ED visits per person-year** vs **0.48** for no-barrier patients, and **0.70 inpatient admits/py** vs **0.22**. Any-ED prevalence is **68% vs 43%**; any-inpatient is **63% vs 35%**.
2. **The signal survives age and sex adjustment.** In a logistic model (`outcome ~ transport + age + sex`, **n = 58,639** screened patients), reporting a barrier carries an **adjusted OR of 3.17 (95% CI 2.93–3.43)** for any ED use and **3.49 (3.23–3.77)** for any inpatient admission. Barrier patients are *younger* (median age 51 vs 61), ruling out an age artifact.
3. **The gap persists inside each chronic disease.** Restricting to patients with hypertension, diabetes, CKD, or AFib index encounters, the 180-day ED-return rate roughly **doubles** for barrier patients (HTN **33% vs 15%**; T2D **40% vs 17%**; CKD **37% vs 21%**; AFib **34% vs 24%**). Barrier patients also visit more often (median 4-day gap vs 8-day), consistent with a more reactive unscheduled pattern.

Caveats

- Only **61,052 of 947,685** patients (~6%) have any transportation answer; screening is non-random, so absolute prevalences should not be projected to the full system.
- **20%** of encounters carry a `PrimaryDiagnosisKey` not found in the diagnosis lookup (known data issue per the Q&A), so the chronic-disease cohort is a lower bound.
- **65%** of patients have no parseable FIPS code, so no geographic model was fit.
- `DepartmentType` is `*Unknown` for 71% of rows, so setting analyses use boolean encounter flags (`IsEdVisit`, `IsInpatientAdmission`, ...).
- Patient journeys are left- and right-censored at the 2022-01-01 / 2025-12-31 window; rates are annualized by observed follow-up.

Implication

A single question — *"Has lack of transportation kept you from medical appointments?"* — identifies a cohort with **~3x odds** of acute-care need, independent of age. This is a high-yield intervention target: reliable rides, co-located pharmacy delivery, tele-visits for chronic-disease follow-up, and case management. Acting on the transport answer that *Stormont Vail already collects* is a concrete next step.

Code and reproducible pipeline (*R + DuckDB*) accompany this write-up as `TeamXX_Code.txt`.